NYC YELLOW TAXI DATA ANALYTICS

Business Insights & Recommendations | May 2026

1 PROJECT OVERVIEW :-

- Project objective: analyze trip demand, recorded trip amounts, fares, distances, pickup/drop-off zones, payment and tipping behavior, vendor activity, airport trips, and data-quality exceptions.
- Technology stack: Python (Pandas) for data profiling and cleaning, PostgreSQL for structured analysis, and Power BI for interactive reporting.
- Scope: May 2026 Yellow Taxi trip data. Results describe this reporting period and should not be generalized to other months without additional data.

1. BUSINESS PROBLEM :-

- The raw trip file contains **40M** of records, making manual review and ad hoc reporting impractical.
- The business needs a consistent view of when and where trips occur, how trip amounts and fares vary, and which records should be reviewed for data-quality concerns.
- The analysis must distinguish recorded trip amounts from profit, compare both counts and rates where appropriate, and avoid unsupported causal conclusions.
- Because the dataset covers one month, it supports within-month daily/hourly analysis but not month-over-month or year-over-year comparisons.

2. DATASET SUMMARY :-

Dataset item	Project detail
Reporting period	May 2026 — NYC Yellow Taxi
Raw trip table	4,090,836 rows × 20 source columns
Final cleaned trip table	4,050,862 rows
Rows removed	39,974 (0.98% of raw rows)
Taxi zone lookup	265 rows initially; 263 rows after dropping rows with missing lookup fields
Analysis tools	Python, PostgreSQL, Power BI

- The main fact table is stored in PostgreSQL as `yellow_tripdata`; the zone reference table is `taxi_zone_lookup`.
- The trip table is the source for trip-level measures; the zone lookup provides Borough, Zone, and service-zone context for pickup/drop-off analysis.
- The project includes analyst-created quality fields and flags in addition to original TLC fields. These should be documented separately from source columns.

3. Python — Data Cleaning & Feature Engineering :-

- Profiled row counts, columns, missing values, data types, duplicates, and suspicious numeric/date values before analysis.
- Checked negative monetary values, zero trip distances, non-positive trip durations, invalid dates, extreme fare/distance values, cash trips with recorded tips, total-amount component mismatches, and invalid location IDs.
- Converted passenger count and rate code to nullable integer types where appropriate; standardized selected missing or special values and derived trip duration and distance in kilometres.
- Flagged outlier and reconciliation issues; set selected invalid values to missing rather than dropping the entire trip when retaining the trip record was appropriate.
- The cleaned dataset contains 4,050,862 trips. The 39,974 removed rows represent approximately 0.98% of the original 4,090,836 rows.

4. SQL (PostgreSQL) — Analysis and KPIs :-

- Loaded the cleaned trip data into `yellow_tripdata` and the location reference into `taxi_zone_lookup` for repeatable business analysis.
- SQL analysis supports trip counts by pickup/drop-off zone, time-based demand, revenue/amount analysis, payment patterns, vendor comparisons, airport-related activity, and exception review.
- Use a LEFT JOIN when all fact-table trips must remain visible even if a location lookup match is missing; use the lookup fields to group results by Borough and Zone.
- Validate Power BI measures against SQL using the same filters, date scope, NULL handling, and KPI definitions before publishing results.

5. Power BI Dashboard :-

- Executive Overview — headline KPIs and a summary of trip activity.
- Demand Analysis — trip volume by time and demand patterns.
- Zone Analysis — pickup/drop-off geography and zone-level comparisons.
- Revenue and Fare Analysis — recorded trip amounts, fare patterns, distance relationships, and geographic comparisons.
- Payment and Tips — payment-method mix and tipping patterns.
- Vendor and Airport — vendor activity, airport-related trips, and rate-code/service patterns.
- Exception Monitoring — records flagged for unusual time, distance, fare, tip, total-amount, or location conditions.
- A tooltip page supports contextual detail. Slicers should make date, geography, vendor, and exception review straightforward.

6. KEY INSIGHTS :-

- **Cleaning reduced the fact table by 0.98%.** The row count decreased from 4,090,836 to 4,050,862; 39,974 rows were removed. This is a cleaning outcome, not a measure of business loss.
- **Zero-distance records were a material audit category.** The initial audit recorded 111,259 zero-distance rows (2.73% at that audit stage). The cleaning rule set zero trip distances to missing instead of treating them as valid measured distances.
- **Non-positive trip durations required handling.** The initial audit flagged 52,063 records with duration at or below zero. Those duration values were set to missing during cleaning.
- **Fare and distance extremes were reviewed separately.** The audit flagged 31,982 fare outliers and 68 distance outliers under the project rules. These are audit-stage counts; categories can overlap and should not be added together.
- **Total-amount reconciliation needs careful interpretation.** The audit-stage component check flagged 550,780 total-amount mismatches. Under the documented rule, affected `total_amount` values were set to missing; this count should not be presented as confirmed fraud.
- **Additional exception checks found targeted issues.** The audit identified 25 cash-payment records with a recorded tip and 25,019 invalid-location records before invalid locations were removed. These categories may overlap with other flags.
- **The time window limits trend claims.** May 2026 supports daily and hourly analysis within the month. A reliable MoM or YoY comparison requires comparable data from additional months or years.

7. BUSINESS RECOMMENDATIONS :-

- **Use both counts and rates.** When comparing boroughs or vendors, show exception counts alongside exception rates. Higher-volume groups can have more flagged trips simply because they handle more trips.
- **Make exception categories independent.** A trip can have more than one issue. For the Exception Monitoring slicer, use a disconnected Exception Type table with a dynamic measure or a normalized exception table; avoid a priority-based column that hides secondary flags.
- **Document monetary KPI logic.** Clearly define whether Total Revenue uses `total_amount` only or falls back to `expected_total` when `total_amount` is missing. Label it as recorded trip amount/revenue proxy where appropriate, never profit.
- **Reconcile headline measures.** Cross-check Total Trips, Total Revenue, average trip amount/fare, airport-trip share, and exception counts between Python, SQL, and Power BI using consistent rules.
- **Investigate patterns, not just outliers.** Review repeated amount mismatches, unusual fares, extreme distances, and timing issues by date, zone, and vendor. Treat flags as review signals, not proof of fraud or misconduct.
- **Use the dashboard for targeted operational review.** Use hourly demand and high-volume zone views to inform dispatch questions; validate patterns over more than one month before making durable staffing or service decisions.
- **Expand the comparison window.** Add more months of comparable Yellow Taxi data before publishing MoM/YoY changes, seasonality conclusions, or claims that a pattern is persistent.

Data integrity note: Audit-stage counts above come from different steps in the cleaning workflow, may overlap, and must not be summed. Exact final revenue totals, leading zone rankings, vendor rankings, and final exception rates were not independently extracted from the PBIX visual outputs during this review; they are intentionally omitted rather than estimated.